

compared to LCD screens. This means that OLED displays require less power and also can be made very thin. Without any other source of illumination, OLED screens can display bright images that are viewable from almost any angle.

Like LCDs, OLEDs also come in passive-matrix and active-matrix flavours.

Passive-matrix OLEDs are made up of a matrix of electrically-conducting rows and columns making pixels. Between the rows and the columns are the organic layers. On the other side is the substrate, the material which gives the electricity. The more current applied, the brighter the display.

In Active-matrix OLEDs, there is a TFT back plate instead of rows and columns. This controls the brightness of each pixel. There are two TFT arrays per pixel: one to start and stop the charging of the capacitor, and one to provide a constant electrical current to the pixel. Active-matrix OLEDs consume much lower power than passive-matrix ones.

OLEDs have the following advantages over LCD or plasma displays:

New-age displays: OLEDs can bring in new types of displays, like ultra-thin, flexible or transparent displays.

Power efficiency: Since OLED screens do not require a backlight, they require much less power compared to other types of displays. This makes OLED a far better choice for portable devices. It also makes them much more environmentally friendly.

Greater brightness: Due to their high contrast and luminance, OLED screens are brighter and have a fuller viewing angle. **Greater clarity:** Since there is no intervening liquid crystal material that limits colour vibrancy, OLED screens offer better colour reproduction.

Lifelike motion: OLED pixels can turn on or off very fast, and